

Model Performance Comparison and Application in Cattle Biometric Identification

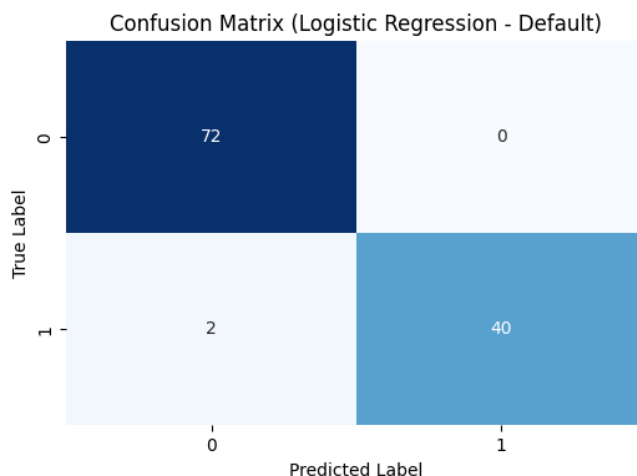
Introduction

Supervised machine learning models such as Logistic Regression and Support Vector Machines (SVM) are widely used for classification tasks in various fields. This study compared the performance of Logistic Regression and SVM models with different parameter configurations on a binary classification dataset. Evaluation metrics including accuracy, precision, recall, F1-score, and confusion matrices were used to assess performance. Finally, the practical applications of Logistic Regression and SVM in precision livestock farming (PLF), specifically biometric cattle identification, are discussed.

Results Comparison

Logistic Regression (L2 Regularization, $C = 0.6$)

- Accuracy: **0.98**
- Precision: Class 0 = 0.97, Class 1 = 1.00
- Recall: Class 0 = 1.00, Class 1 = 0.95
- F1-score: Class 0 = 0.99, Class 1 = 0.98
- Confusion Matrix:

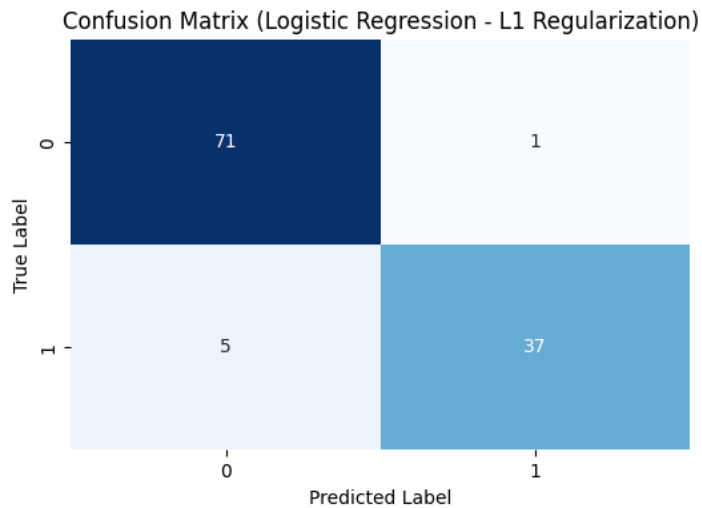


- **Observation:** Excellent balance, only two misclassifications.

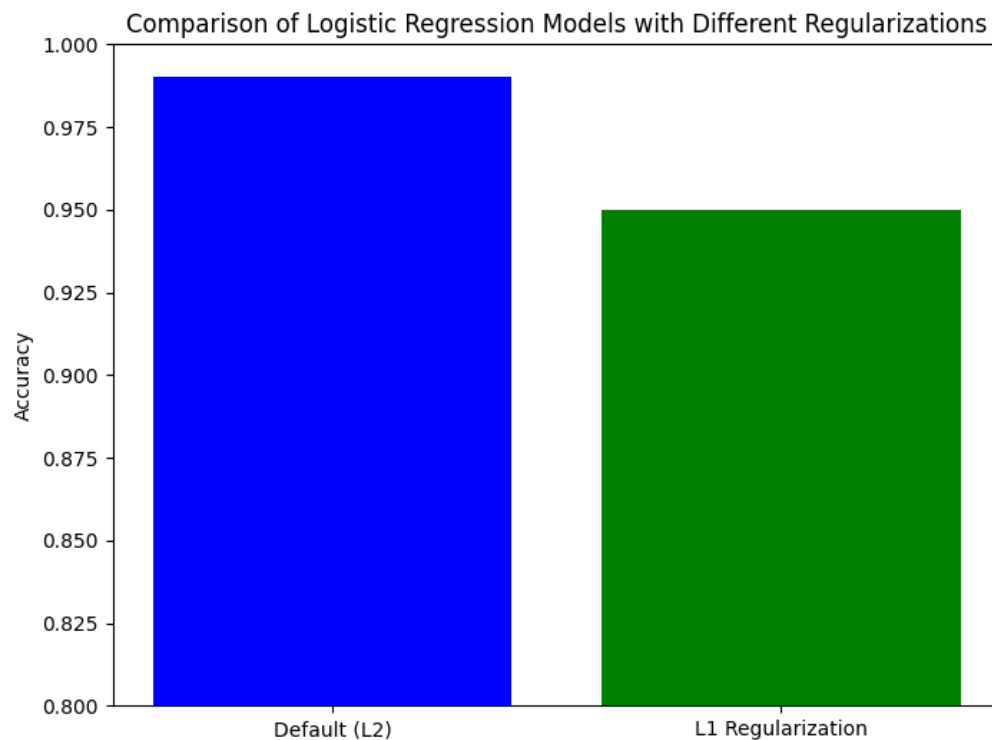
Logistic Regression (L1 Regularization)

- Accuracy: **0.95**

- Precision: Class 0 = 0.93, Class 1 = 0.97
- Recall: Class 0 = 0.99, Class 1 = 0.88
- F1-score: Class 0 = 0.96, Class 1 = 0.93
- Confusion Matrix:



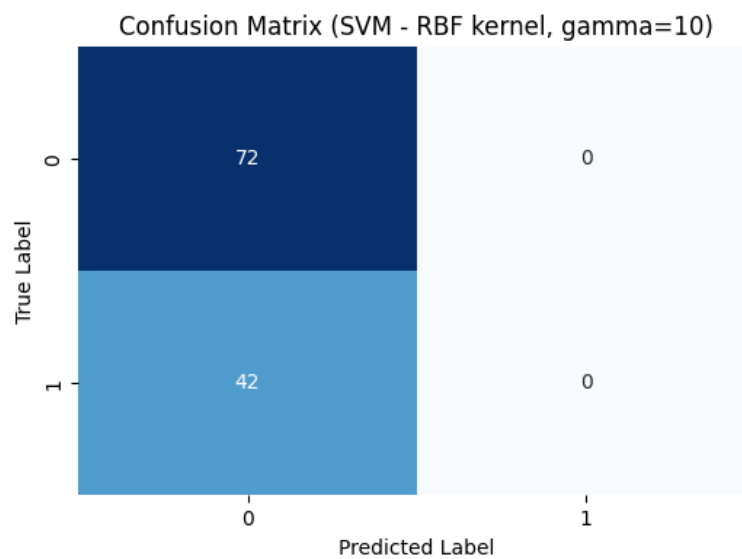
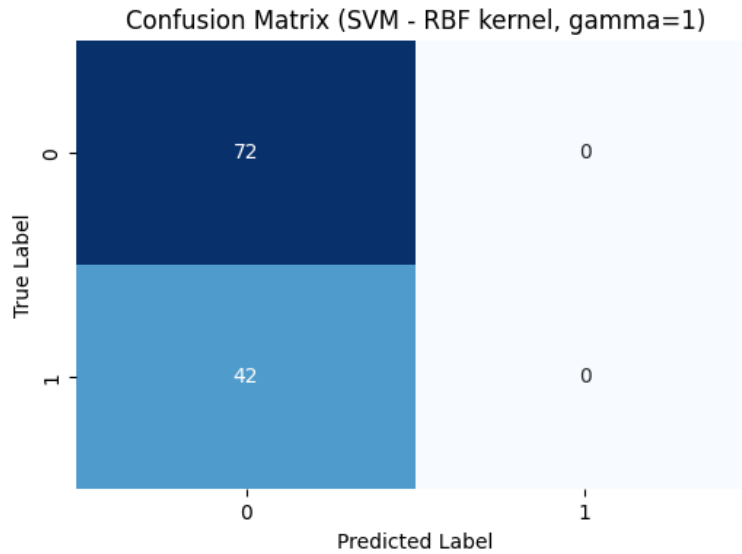
- **Observation:** Slightly worse than L2 due to higher false negatives for Class 1.



SVM (RBF Kernel, gamma = 1 and 10)

- Accuracy: **0.63**

- Precision (Class 1): 0.00 (model failed to classify Class 1)
- Recall (Class 1): 0.00
- Confusion Matrix:

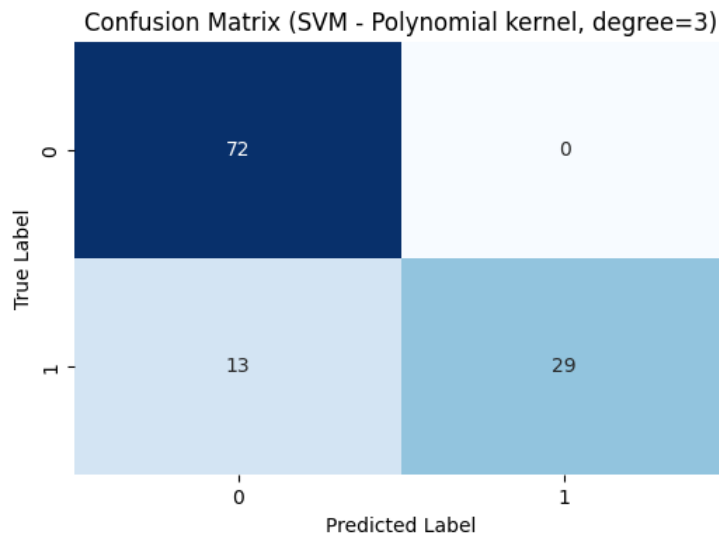


- **Observation:** Very poor performance, strong bias toward Class 0.

SVM (Polynomial Kernel, degree = 3)

- Accuracy: **0.89**
- Precision: Class 0 = 0.85, Class 1 = 1.00
- Recall: Class 0 = 1.00, Class 1 = 0.69

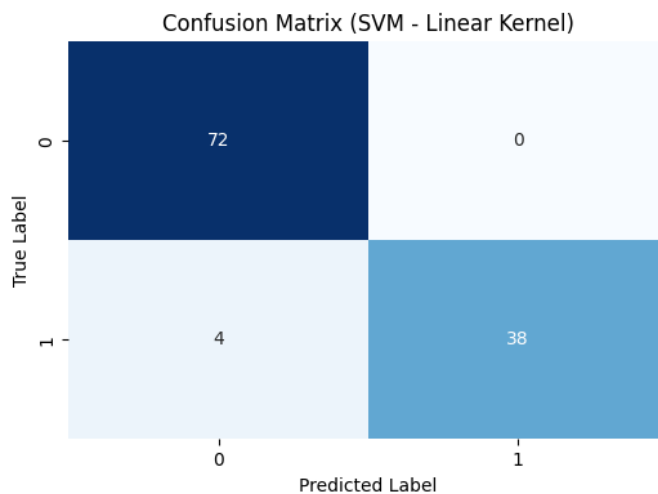
- F1-score: Class 0 = 0.92, Class 1 = 0.82
- Confusion Matrix:



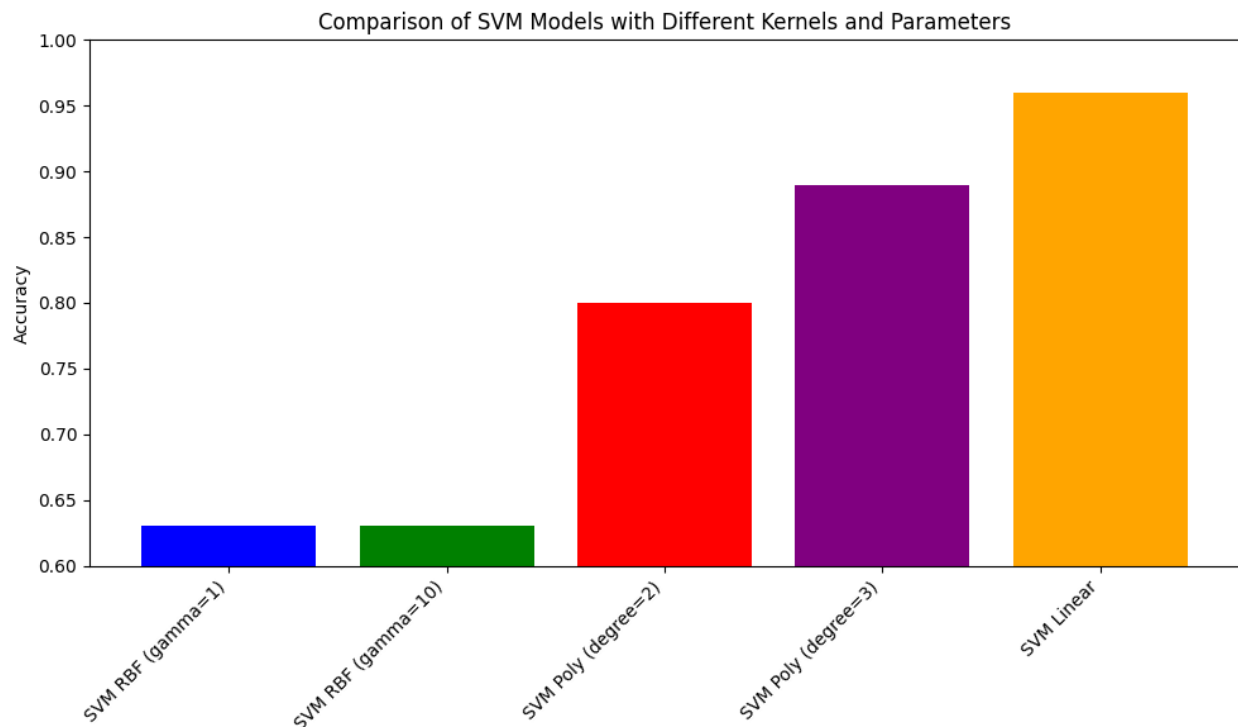
- **Observation:** Better than RBF but lower recall for Class 1.

SVM (Linear Kernel)

- Accuracy: **0.96**
- Precision: Class 0 = 0.95, Class 1 = 1.00
- Recall: Class 0 = 1.00, Class 1 = 0.90
- F1-score: Class 0 = 0.97, Class 1 = 0.95
- Confusion Matrix:



- **Observation:** Strong performance, close to Logistic Regression (slightly weaker Class 1 recall).



Discussion

The results clearly show that **Logistic Regression with L2 regularization ($C=0.6$)** achieved the best performance, with an accuracy of 98% and very balanced precision, recall, and F1-scores. **SVM with a linear kernel** performed nearly as well, with 96% accuracy.

In contrast, SVM with RBF kernels ($\gamma = 1, 10$) failed completely, while polynomial kernel SVM performed moderately well but struggled with recall for Class 1. This highlights that kernel and parameter choices have a significant impact on SVM's effectiveness.

Application in Cattle Biometric Identification

Background

In precision livestock farming, individual cattle identification is critical for monitoring health, feed intake, and overall management. Traditional methods such as RFID or ear-tags have limitations due to loss, tampering, or visibility issues. Biometric approaches (e.g., muzzle patterns, head whorls, ear-tag OCR, facial recognition) provide a more robust alternative.

Predictor Variables (X)

- Image-based biometric features extracted using computer vision (YOLOv11x + CNN embeddings).
- Muzzle texture patterns.
- Head whorl orientation and position.
- Ear-tag number (digit recognition).
- Face shape descriptors.

Dependent Variable (Y)

- **Cattle ID (multi-class classification):** Assign an image to one of N known cattle.
- **Match vs. Non-Match (binary classification):** Verify if an image belongs to a specific cow.

Using Logistic Regression

- Logistic Regression can be used for **verification tasks** (binary classification).
- Example: Predict whether extracted biometric features (e.g., muzzle texture) match a target cow.
- Pros: Fast, interpretable, less computationally intensive.
- Application: On Jetson Orin Nano, Logistic Regression could run locally for quick verification of an animal's ID.

Using SVM

- SVM, particularly with a **linear kernel**, is effective for **multi-class classification**.
- Example: Classify a cattle image into one of many possible IDs based on biometric features.
- Pros: Works well in high-dimensional feature spaces (e.g., CNN embeddings).

- Application: On edge devices, SVM can handle multi-class recognition of cattle from images captured in feedlots.

Conclusion

- The best-performing model in this study was **Logistic Regression with L2 regularization (C=0.6)**, achieving 98% accuracy.
- **SVM with a linear kernel** also performed strongly (96% accuracy) and is suitable for multi-class scenarios.
- In cattle biometric identification research, **Logistic Regression** is well-suited for verification tasks, while **SVM** can be applied to large-scale multi-class identification.
- Together, these models can support real-time cattle recognition pipelines, improving management and monitoring in precision livestock farming.

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1st October 2025